Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/21/2017
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of P21/CDKN1a Pathway in Microgravity-Induced Bone Tissue Regenerative Arrest - A Spaceflight Study of Transgenic P21/CDKN1a Null Mice in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	E.Almeida@nasa.gov	**Fax:**	FY 650-604-3159
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1772
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 236-7		
PI Address 2:	Building 236 Room 217		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	12/31/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2020 per F. Hernandez/ARC (Ed., 6/20/17)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Blaber, Elizabeth Ph.D. (NASA Ames Research Center) Globus, Ruth Ph.D. (NASA Ames Research Center) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tissue degeneration and the failure of normal tissue regenerative processes in microgravity are increasing concerns for sustaining life in space as the duration of future flight missions increases. Specifically our laboratory has identified normal adult stem cell-based tissue regenerative processes, such as the formation of new bone, cartilage, immune cells, and blood from bone marrow mesenchymal stem cell precursors as being particularly sensitive to the stresses of mechanical unloading in microgravity. The cellular mechanisms activated by microgravity unloading in mouse bone during spaceflight are not completely understood, but they include induction of oxidative stress and the inhibition of differentiation of marrow mesenchymal stem cells, via activation of p21-mediated cell cycle arrest in proliferative osteprecursor cells on the bone surface. The identification of p21 mRNA and protein as being highly up-regulated by spaceflight in mice is a seminal finding, because this gene is a master regulator of tissue regeneration that when knocked out allows for mammalian limb regeneration similar to that found in amphibians. The role of p21 as a key player in regulating the dynamics of bone formation and regeneration has also been established in our laboratory's studies of bones from p21 null mice, which show unrestrained growth with significantly increased bone mineral content and robust structure, and in in-vitro osteoblastogenesis from bone marrow stromal cells that show greatly increased proliferation and very early mineralization. Because we have established in multiple spaceflight and ground experiments that the p21 pathway is a key molecular mechanism of inhibiting tissue regenerative processes specifically activated in microgravity, we now seek to fly p21 null mice in microgravity to test the hypothesis that this gene is required for the arrest of regenerative bone formation observed in microgravity. Specifically we propose to fly five transgenic mice and five wildtype mice on the International Space Station (ISS) for a period of about 30 days with administration of the cell proliferation marker BrdU at the end of the experiment. This strategy will allow us to determine the extent of regenerative cell proliferation in adult stem cells after exposure to microgravity in the presence and absence of the gene hypothesized to mediate regenerative arrest in space. We propose to study tissues including the pelvis, femur, and tibia using various approaches including RT-qPCR, gene arrays for gene expression analysis, immunolocalization of BrdU in histological sections for analysis of proliferation, and microCT and nanoCT for bone structural analysis. In summary the study we propose directly addresses NASA's and the National Research Council (NRC) Decadal Survey call to study bone degenerative/regenerative processes in microgravity, including its molecular mechanisms, using transgenic mouse models. When completed, these studies will provide molecular mechanistic evidence for how p21, one of the most important master regulatory genes of stem cell regenerative heath, is modulated by microgravity, and could be the basis for development of preventive treatments designed to counter tissue degeneration in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Recent advances in the scientific understanding of stem cells and their role in tissue regeneration created a new field of regenerative medicine that holds great promise both in the treatment as well as prevention of many degenerative diseases often associated with aging. In short duration human spaceflight, when the body is weightless and unloaded in microgravity, multiple tissue degenerative conditions occur, such as muscle atrophy, bone loss, immune deficiencies, and others that may have a deficit in stem cell-based tissue regeneration in common. Likewise, on Earth, when we increase mechanical loading of the body with exercise we observe an increase in adult stem cell-based tissue regeneration, such as in bone strengthening, muscle mass increase, and increased production of immune cells and red blood cells from hemotopoeietic stem cell progenitors. In total, spaceflight and ground experimental evidence from human and animal studies now strongly suggests that stem cell-based regenerative health is regulated by mechanical loading of the body, increasing with exercise and load, and decreasing with inactivity and unloading as in microgravity. Our proposed experimental work in microgravity flight experiments and mechanical unloading models addresses the molecular, cellular, and tissue aspects of the hypothesis that gravity mechanical loading regulates regenerative health. The novel results we obtained identify the transition between stem cell progenitors and differentiated cells as a likely cellular process regulated by mechanical loading and specific molecular candidates such as the p21/CDKN1a pathway induced in microgravity as inhibitory of stem cell-based tissue regeneration in bone. This work is highly significant to NASA space exploration and medicine on Earth because it holds the promise of greater scientific understanding of how to maintain and monitor health during future voyages of space exploration, and solidly establishes Earth gravity as a major shaper of human life through promoting tissue regenerative health.
	Because of limited spaceflight opportunities to conduct the research outlined in the NASA Grant "The Role of CDKN1a/p21 Pathway in Microgravity-Induced Bone Tissue Regenerative Arrest – A Spaceflight Study of Transgenic CDKN1a/p21-Null Mice in Microgravity" and following promising initial definition work with HU (hindlimb unloading) and characterization of the bone phenotype, we proposed to Space Biology a modification to our existing grant to achieve the same science aims originally proposed but using an expanded set of approaches, also including more use of the microgravity ground model of Hind-limb Unloading (HU), and the analysis of recently awarded Rodent Research 1 mouse bone tissues exposed to microgravity for 37 days, and existing Bion M1 tissues (exposed to microgravity for 30 days) to study CDKN1a/p21 status and its correlation to bone regenerative processes. This request was approved in early 2016 and is the basis for the current report. Specifically we restructured the period of grant performance from 3 years to two separate 2-year phases, while still maintaining the same total cost. These include an initial 2-year ground research phase starting in fiscal year 2016, following initial definition, and a subsequent 2-year flight phase to start when a flight opportunity becomes available. In the initial new ground research phase starting March 2016 we began addressing all the aims (1a, 1b, and 1c) using the mouse hind-limb unloading HU model of microgravity and preparing samples for verifying CDKN1a/p21 status in microgravity bone samples that we now have from a RR1 tissue-sharing award and from the previous spaceflight experiments on Bion M1. The performance of HU experiments was already proposed and is part of the original grant, for ground verification of the flight experiment procedures and science definition, and has now been expanded to a greater number of analyses, on both males and female mice at 15 and 30 days of duration. Full scale breeding of p21 null mice started early in 2016 and led to the first full set of HU experiments in p21null and wt male mice with and without HU at 15 and 30 days during the summer of 2016. Although we had already conducted a small scale pilot HU experiment early on during the definition phase of the grant, the current experiments were now conducted simultaneously with larger numbers of animals, BrdU injections to label proliferating osteoprogenitor cells in bone and in bone marrow, and with calcein injections at the beginning and end of unloading experiments, to label the extent and localization of new bone mineral deposition. Besides providing an important tool for measurement of bone formation with and without unloading in the presence and absence of the cell cycle progression inhibitor CDKN1a/p21, these experiments are also crucial in refining spaceflight procedures in preparation for experiment verification, as the timing of injections and duration of labeling often determines the success of bone formation and cell proliferation measurements. A second set of HU experiments will start shortly in January 2017, with female mice, essentially repeating the studies

Task Progress:	already conducted in males to address sex-specific responses of bone to unloading. Tissues from the first large set HU experiments with male mice mice were already used for microCT imaging at 6 micron resolution which was recently concluded. MicroCt reconstruction and analysis is now underway and will be followed by more detailed imaging at 600 nm resolution using a recently acquired nanoCT instrument, which we expect will deliver single cell/osteocytic lacunar and bone mineral structure resolution on bone remodeling dynamics at sites of regeneration in the p21 and wt backgrounds with and without load. In addition, we have also initiated analysis of gene expression measured by Affymetrix gene arrays using RNA extracted from bone and bone marrow tissue samples, for specific analysis of the CDKN1a/p21 pathway and potential involvement of the CIRBP gene. These include Bion M1 samples already processed, male HU experiment samples currently being processed, with RR1 and upcoming female HU experiments still to be processed. Gene expression analysis and processing underway includes data normalization, grouping, and qc for significance and gene ontology analysis using GeneSpring software. In addition analysis of alterations in gene expression pathway is also being conducted using KEGG and GSEA. For histological analysis, thus far, we have processed the femoral head and proximal femur of bones from male HU experiments and Bion M1 mice and conducted bone marrow stromal cell differentiation assays for mineralized tissue and hematopoietic progenitor differentiation into osteoclasts. BrdU and calcein labeling detection experiments have not yet been conducted but are planned for early 2017. Finally, we initiated in-vitro studies of cell cultures subjected to uniaxial loading and also started to investigate gap junction and integrin adhesion signaling in the context of p21 status, and how it may contribute to sensing and responding to gravity mechanotransduction. Our preliminary results from microCT analysis of BionM1 femurs revealed 31% decrease in bone volume ratio, a 14% decrease in trabecular thickness, and a 20% decrease in trabecular number in the femoral head of space-flown mice. Furthermore, high-resolution MicroCT and immunohistochemical analysis of spaceflight tissues revealed a severe disruption of the epiphyseal boundary, resulting in endochondral ossification of the femoral head and perforation of articular cartilage by bone. This suggests that spaceflight in microgravity may cause rapid induction of an aging-like phenotype with signs of osteoarthritic disease in the hip joint. Analysis of gene expression in bone of mice from Bion M1 also shows a 1.5 fold elevation of CDKN1a/p21 levels relative to ground controls, not as high as the 4-fold previously observed on STS-131 samples, although these samples were recovered many hours post landing and reloading, which may have suppressed p21 levels. Mice from RR1 exhibited significant bone loss in the femoral head but did not exhibit as severe aging and disease-like phenotype observed during BionM1. This may be due to increased physical activity in the RH hardware. Immunohistochemical analysis of the epiphyseal plate and investigation of cellular proliferation and differentiation pathways within the marrow compartment and whole bone tissue is currently being conducted to determine alterations in stem cell-based tissue regeneration, bone loss and CDKN1a/p21 status between these experiments. Likewise, mice from HU experiments also showed significant bone loss in the femoral head at 15 and 30 days but with lesser levels of endochondral ossification and cartilage loss than observed during Bion M1. Our results thus far suggest that inhibition of stem cell-based tissue regeneration and CDKN1a/p21 elevation persists during long-duration spaceflight, with spaceflight femurs from Bion M1 showing the onset of an accelerated aging-like phenotype with signs of osteoarthritic disease evident in the disruption of the epiphyseal boundary and endochondral ossification, as well as evidence for CDKN1a/p21 involvement. These effects appear to be related to a failure of stem cell chondroprogenitors to regenerate, and possibly an imbalance between osteogenesis and chondrogenisis, resulting in extensively remodeled femoral head tissues that may have significant implications for bone and cartilage health following extensive periods of mechanical unloading during long-duration HU or spaceflight.

Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Blaber EA, Cheng-Campbell MA, Almeida EAC. "Femoral head bone loss following short and long-duration spaceflight." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Juran CM, Blaber EA, Almeida EAC. "Mechanotransductive regulation of gap-junction activity between MLO-Y4 osteocyte-like and MC3T3-E1 osteoblast-like cells in three-dimensional co-culture." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Cheng-Campbell MA, Blaber EA, Almeida EAC. "Changes in mouse bone turnover in response to microgravity." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Blaber EA, Almeida EAC. "The role of the CDKN1a/p21 pathway in microgravity-induced arrest of stem cell-based tissue regeneration." International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. , Jun-2016
Abstracts for Journals and Proceedings	Blaber EA, Almeida EAC. "The Severity of Bone and Cartilage Degeneration in the Femoral Head of Mice During 30 days of Microgravity Exposure Is Dependent on Physical Interactions with Their Habitat." International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. , Jun-2016

Abstracts for Journals and Proceedings	Blaber EA, Shirazi-Fard Y, Almeida EAC. "The role of the CDKN1a/p21 pathway in microgravity-induced arrest of stem cell-based tissue regeneration." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Blaber EA, Cheng-Campbell MA, Chen JG, Almeida EAC. "CDKN1a/p21 regulates load-dependent stem cell-based osteoprogenitor differentiation into mineralized tissue." 2015 American Society for Cell Biology (ASCB) Annual Meeting, San Diego, CA, December 12-15, 2015. Molecular Biology of the Cell. 2015 Dec 15;26(25):P1554. http://dx.doi.org/10.1091/mbc.E15-09-0674 , Dec-2015
Abstracts for Journals and Proceedings	Blaber EA, Shirazi-Fard Y, Almeida EAC. "CDKN1a/p21 suppresses osteogenesis and regenerative bone remodeling in an age-dependent manner." 37th Annual Meeting of the American Society for Bone and Mineral Research, Seattle, Washington, October 9-12, 2015. Journal of Bone and Mineral Research. 2015;30 (Suppl 1). , Oct-2015
Abstracts for Journals and Proceedings	Chen A, Chen J, Cheng-Campbell MA, Blaber EA, Almeida EAC. "Effects of Uniaxial Strain on Stem Cells from Wildtype and CDKN1a/p21 Knockout Mice." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	DeMatteo CJ, Cheng-Campbell MA, Chen JG, Shirazi-Fard Y, Blaber EA, Almeida EAC. "Alterations in Stem Cell and Mechanical Transduction Signaling Pathways in Hindlimb Unloaded CDKN1a/p21 Knock-out Mice." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Cheng-Campbell MA, Chen JG, Shirazi-Fard Y, Blaber EA, Almeida EAC. "Role of CDKN1a/p21 in Mouse Bone Turnover during Mechanical Unloading." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Blaber EA, Cheng-Campbell MA, Chen JG, Shirazi-Fard Y, Almeida EAC. "CDKN1a/p21 Alters Osteogenesis and Regenerative Bone Remodeling in a Mechanical Load- and Age-Dependent Manner." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015